

P S Lochan

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Education

New Horizon College of Engineering, BE in Electronics and Communications Sept 2023 – Sept 2027 (Expected)
Minor in Artificial Intelligence and Machine Learning

- GPA: 8.34 / 10.0
- **Relevant Coursework:** Quantum Computing, Machine Learning, Deep Learning, Blockchain Technology, Smart Contracts, Linear Algebra, Probability & Statistics

Experience

QOVA Labs — ML Intern Feb 2026 – Apr 2026

- Built an ask-user style conversational system inspired by Claude-like interaction paradigms.
- Designed and ran internal evaluation pipelines and benchmarks to improve response quality and system performance.
- Worked on iterative refinement loops to enhance model reasoning and usability.

ALPHAI — ML Intern Oct 2025 – Nov 2025

- Trained foundational models for stock movement prediction using novel approaches inspired by natural phenomena (e.g., entropy-based modeling).
- Worked on improving prediction robustness and generalization across varying market conditions.

Founder — **Quantum Collective** 2026 – Present (Community Link)

- Built a fast-growing quantum community (200+ members in 2 weeks) connecting students, researchers, and developers.
- Created an open environment for collaboration, learning, and discussion across all levels of quantum computing.

Quantum Educator — “Quantum with Lochan” (YouTube) (Channel Link)

- Teaching quantum computing from first principles to make the field accessible to beginners.
- Publishing 2 videos per month focused on intuitive understanding and practical learning.

Achievements

- **Winner - IISc Bengaluru × IBM Qiskit Fall Fest 2024** – Developed BB84 Quantum Key Distribution Network Simulator with multi-attacker scenario analysis and real-time web visualizations.
- **Finalist - Cardano Blockchain Hackathon** – Built Subscribe, a privacy-first subscription management platform on Cardano with integrated payment service provider (2.5% commission model).
- **IBM Qiskit Advocate 2025** – Represented the global student quantum community; contributed to Qiskit discussions and developer events. (Advocate ID: 116901)
- Invited to **IBM Quantum Developers Day 2025** [Post] and **NASSCOM AI Confluence 2025** [Post]
- **Smart India Hackathon 2025 (2nd Round Qualified)** – Built an AI-based Rockfall Prediction System for Open-Pit Mines.

Projects

Kanad – Quantum Native Digital Laboratory [Website]

- Developed a web-based quantum chemistry platform enabling real quantum computation through an intuitive laboratory interface.
- Built on governance-driven framework achieving chemical accuracy in 17 evaluations (49x faster) using PhysicsVQE.
- Enabled execution across IBM Quantum, BlueQubit, and IonQ with configurable solvers and molecular builders.
- Supported 20+ molecular analysis properties including bond-specific Hamiltonians, molecular dynamics, and environmental effects.

- **Tech:** Quantum Computing, VQE, SQD Solvers, Molecular Simulation, Web Technologies
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BB84 Quantum Key Distribution Network Simulator

[GitHub] — *Hackathon Winner*

- Built complete web-based implementation of BB84 protocol with multi-attacker scenario analysis and real-time visualizations.
 - Developed Flask REST API backend with Qiskit for quantum simulation and React frontend with interactive controls.
 - Implemented 5 attack scenarios including intercept-resend attacks with statistical error detection (QBER threshold: 11%).
 - **Tech:** Python, Qiskit, Flask, React, Vite, Matplotlib, NumPy
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AI Dreams – Emotional Companion with Dream Cycles

[GitHub]

- Developed AI system with circadian rhythm: awake → drowsy → dreaming → waking states, processing memories during sleep cycles.
 - Implemented real-time emotion detection (7 states) with confidence scoring and therapeutic audio system (30+ tracks).
 - Built 3D neural network visualization using React Three Fiber showing dream symbols and memory processing.
 - Created personality evolution system with bounded parameter shifts (± 0.15) based on dream-processed memories.
 - **Tech:** React, TypeScript, Zustand, LangGraph, OpenRouter API, Three.js, Framer Motion
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Subscribe – Cardano Subscription Management Platform

[GitHub]

- Built privacy-first subscription management platform on Cardano with integrated Payment Service Provider (2.5% commission).
- Developed Plutus smart contracts for subscription vault and automated payment routing on Cardano L1.
- Implemented complete dashboard with analytics, calendar view, and merchant revenue tracking across 7+ pages.
- **Tech:** Next.js 14, TypeScript, Tailwind CSS, Plutus (Haskell), Express.js, Recharts

Technical Skills

- **Languages:** Python, C
- **AI/ML Frameworks:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Hugging Face Transformers, LangGraph
- **Quantum Computing:** Qiskit, PennyLane, Quantum Circuits, QAOA, VQE, BB84 Protocol
- **Blockchain:** Cardano, Plutus Smart Contracts, Web3
- **Tools & Technologies:** Git, Docker, Linux CLI, VS Code, Jupyter, React, Next.js, Flask, Express.js, Vite, Tailwind CSS
- **Data Visualization:** Matplotlib, Recharts, React Three Fiber, D3.js
- **Areas of Interest:** Quantum Computing, Quantum Machine Learning, Deep Learning, Blockchain, Cryptography, AI Systems